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Program EH User's Manual Version 3

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P1

MANUALS / EH / COMPUTER PROGRAMS

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Pag. 2

Resumen/Summary

The purpose of this Technical Report is to define the way to use the EH program.
The EH program calculates the stress and strain distribution around a circular hole in an infinite orthotropic plate.

This new version of the program includes a graphical windows user interface based in Visual Basic language.

WARNING

Applicability: holes in homogeneous orthotropic flat plates
subject to any combination of in-plane loads (NOT valid for
non symmetric laminates)

Method: ESDU85001

Note to be included in future revisions of user's manual (G.B.L. 16/9/2015)

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1 Technical sheet of the program

Technical Note **ME-T-NT-100006**PROGRAMREFERENCEDATE

EH

2010

OBJECTIVE

This program performs the calculation of the stress and strain distribution around a circular hole in an infinite orthotropic plate.

LANGUAGECOMPUTER

Fortran 77

Sun Solaris

Visual .NET v.9.0

Windows PC's

NUMBER OF LINES (APROX.)EXECUTION MODE

4000

Interactive

AUTHORVERSION

Álvaro García Herrero

3.0

ADDITIONAL DOCUMENTS

Technical Report: NT-00-ECE-059 A. Theoretical Manual of the EH Program. V1.1.

Technical Report: ME-T-NT-100007 Programa EH. Manual del Programador.

2 Introduction

This report makes up the User's Manual for the EH Program, which has been developed by EADS CASA from 1993 to date.

This program performs the calculation of the stress and strain distribution around a circular hole in an infinite orthotropic plate, with the following particularities:

- Allows the user to create a laminate plate, using up to 100 different materials and 500 layers.
- The program calculates the stresses and strains for up to 1000 load cases.
- Moreover, EH program calculates the laminate stiffness matrix and the reserve factors for each load case and ply.
- The results around the hole are given in intervals of 5 degrees.
- EH program, in its windows version, allows the user to plot the results and export them into an image file.

Differences between versions:

Version 1.1:

Initial version developed in Fortran 77.

Version 2.2:

The program is adapted to the new Sun Solaris computer.

Version 3.0:

Now, the user can choose the language for screen dialogs and for output files, between Spanish and English.

The format of the general input data file has some little differences from previous version. Now the data are grouped in the same way as for other applications.

The format of the results is practically the same, with exception of the language.

The program source is restructured to allow the creation of a Dynamic Link Library DLL for Windows operating system.

In parallel with this, a Windows application has been developed as Graphic User Interface GUI for the program EH, adapted to the new platform Microsoft .NET Framework. This application uses the same input files and provides the same output results. The analysis is performed by calling to the DLL (eh.dll), compiled from the same source as the official FORTRAN version.

FORTTRAN source code is redistributed in different files without additional changes in the code.

3 Execution of the program

Prior to the program execution, it is necessary to create the input data file, as described in section 4. The program can be then executed by typing its **full path name**.

If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table), it can be executed by typing the command:

`$eh`

The program asks for the language to use and the name of the input file, as shown in the following example:

```
Idioma / Language:
=====
 1 ... ESPAÑOL
 2 ... ENGLISH
Idioma / Language: 2
Name of the input file: Example.dat
```

The CPU time for calculation depends on the laminate and the number load cases to be considered, being a few seconds as much.

It is possible to execute the program with a single command, including the input filename, the output results filename and the language as parameters. The above example can be executed as follows:

`$eh Example.dat Results.txt 2`

4 Input data file

In this chapter, it is described the structure of the input file, which is required for the correct execution of the EH program.

This file contains all the necessary data that defines the problem. It basically contains four data groups, all them are necessary. They will be numbered from 1 to 4 and can appear in any order.

All the numerical data contained in this file will be read in a free format. The variables will be separated by one or more blank spaces. The numbering of each group will be included in the first column.

Figure 1 below shows the notation and sign criteria defined in the EH program.

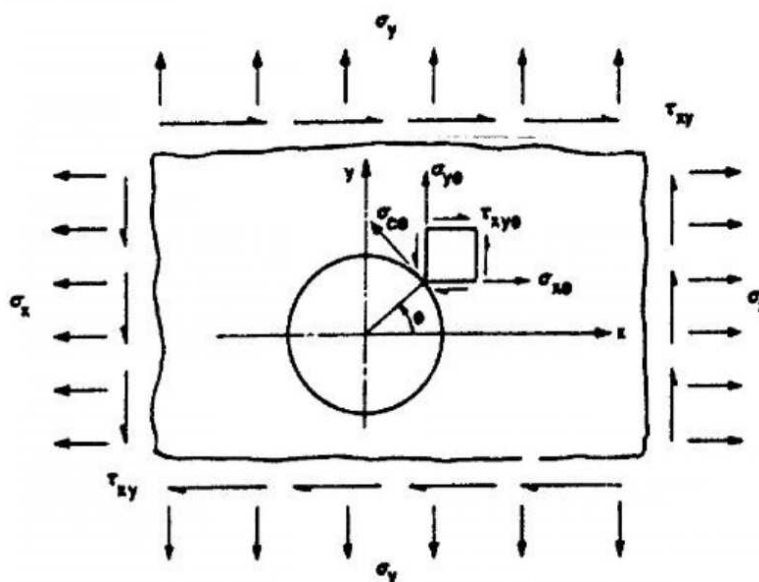


Figure 1: Notation and sign criteria

Listed below are the contents of each data group:

- **GROUP 1: Title of the problem.**

Four lines, containing the title of the problem to be solved, which will appear at the header of the output file:

Line 1 -----> 1 First line of the title

Line 2 -----> Second line of the title

Line 3 -----> Third line of the title

Line 4 -----> Fourth line of the title

- **GROUP 2: Material data.**

Line 1-->2 MATERIAL DATA

Three different ways of material data introduction can be used:

1. Ply properties: In this case, the structure of this group is the following one:

Line 1-->1 Ply properties

[Nm lines] Nm Mat(1) Ex(1) Ey(1) Gxy(1) $\mu_{xy}(1)$ h(1) $\Theta(1)$ et(1) ec(1)

Mat(2) Ex(2) Ey(2) Gxy(2) $\mu_{xy}(2)$ h(2) $\Theta(2)$ et(2) ec(2)

...
...

Mat(Nm) Ex(Nm) Ey(Nm) Gxy(Nm) $\mu_{xy}(Nm)$ h(Nm) $\Theta(Nm)$ et(Nm) ec(Nm)

Where: Nm Number of different materials

Mat(i) Number of material

Ex(i) Elastic Modulus of material i in x-direction

Ey(i) Elastic Modulus of material i in y-direction

Gxy(i) Elastic Shear Modulus of material i in xy-direction

$\mu_{xy}(i)$ Poisson coefficient of material i in xy-direction

h(i) Thickness of material ply i

$\Theta(i)$ Orientation angle of the ply i

et(i) Allowable tension strain of material i

ec(i) Allowable compression strain of material i

2. Stiffness matrix: In this case, the structure of this group is the following:

Line 1-->2 Stiffness matrix

A11 A12 A22 A33 H

Where: A11 Stiffness matrix element: row 1, column 1

A12 Stiffness matrix element: row 1, column 2

A22 Stiffness matrix element: row 2, column 2

A33 Stiffness matrix element: row 3, column 3

H Thickness of the plate

3. Mean plate properties: In this case, the structure of this group is the following:

Line 1-->3 Mean plate properties

E_{xx} E_{yy} G_{xy} ν_{xy}

Where: E_{xx} Elastic Modulus of the plate in x-direction

E_{yy} Elastic Modulus of the plate in y-direction

G_{xy} Elastic Shear Modulus of the plate in xy-direction

ν_{xy} Poisson coefficient of the plate in xy-direction

- **GROUP 3: Lay-Out.**

This group will be required only when the Material data have been introduced using option number 1 (Ply properties). The structure of this group is the following one:

Line 1-->3 LAY-OUT

NPly Ply(1) Ply(2) Ply(3) Ply(4) Ply(i-1)

Ply(i) Ply(i+1) ... Ply(NPly-1) Ply(NPly)

Where: NPly Number of plies

Ply(i) Material number of ply i

- **GROUP 4: Loads.**

This group will contain the applied loads, and the structure of this group is the following one:

Line 1-->4 LOADS

NLoads F_x(1) F_y(1) F_{xy}(1)

F_x(1) F_y(1) F_{xy}(1)

... ...

F_x(NLoads) F_y(NLoads) F_{xy}(NLoads)

Where: NLoads Number of load cases

F_x(i) Applied stress in x-direction for load case i

F_y(i) Applied stress in y-direction for load case i

F_{xy}(i) Applied shear stress in xy-direction for load case i

5 Output results

5.1 Output file

The EH program creates one printable file with all the results of the analysis carried out.

This file includes the following information:

- **Stiffness matrix:**

If the material input data is given as option 1, the results file contains the calculated stiffness matrix.

- **Plate mean properties:**

The results file also contains the resultant mechanical mean properties of the plate.

- **Strain/Stress distribution and Reserve Factors:**

For each load case, the file contains the calculated stresses and strains, showing the values in 5 degrees interval around the hole. Printed values are:

THT	Angle from the x-axis (counter clockwise)
SIGMA	Stress in the direction of THT angle
XSTRESS	Stress in x-axis direction
YSTRESS	Stress in y-axis direction
XYSTRESS	Stress in xy-plane direction
XSTRAIN	Strain in x-axis direction
YSTRAIN	Strain in y-axis direction
XYSTRAIN	Strain in xy-plane direction
ECIRCUM	Strain in circumferential direction
ERAD	Strain in radial direction
ERTHETA	Strain in circumferential-radial plane direction

Examples of the output file are provided in chapter 7.

5.2 Error messages

When the program detects an error in the input data, it writes the corresponding message in the console and stops the execution. The error messages that might appear are listed below:

Error messages. English output:

```
ERROR: The input file cannot be opened'  
ERROR: Bad input data in input file'  
ERROR: Group 1 missing in input file (Title)'  
ERROR: End of file while reading group 1 in input file (Title)'  
ERROR: Group 2 missing in input file (Material data)'  
ERROR: End of file while reading group 2 in input file (Material data)'  
ERROR: Group 2 is defined badly (Entry type of material data)'  
ERROR: Too many material types, exceeding the program capability'  
ERROR: Group 3 missing in input file (Lay-Out)'  
ERROR: End of file while reading group 3 in input file (Lay-Out)'  
ERROR: Group 3 is defined badly (Lay-Out)'  
ERROR: Too many plies in the lay-out, exceeding the program capability'  
ERROR: Group 4 missing in input file (Loads)'  
ERROR: End of file while reading group 4 in input file (Loads)'  
ERROR: Group 4 is defined badly (Loads)'  
ERROR: Too many load cases, exceeding the program capability'  
ERROR: Division by zero in matrix QQ ( $E_x - E_y \cdot V_{xy}^2$ )'  
ERROR: Null value of A11'  
ERROR: Null value of A22'  
ERROR: Null value of H'  
ERROR: Null value of EX'  
ERROR: Null value of EY'  
ERROR: Null value of GXY'  
ERROR: Output file cannot be opened'
```

Error messages. Spanish output:

ERROR: No se puede abrir el fichero de entrada'

ERROR: Datos erróneos en el fichero de entrada'

ERROR: Grupo 1 ausente en el fichero de entrada (Título)'

ERROR: Fin de fichero mientras se leía el grupo 1 en el fichero de entrada (Título)'

ERROR: Grupo 2 ausente en el fichero de entrada (Datos de materiales)'

ERROR: Fin de fichero mientras se leía el grupo 2 en el fichero de entrada (Datos de materiales)'

ERROR: Grupo 2 mal definido (Tipos de entrada de datos de materiales)'

ERROR: Demasiados tipos de materiales, excediendo la capacidad del programa'

ERROR: Grupo 3 ausente en el fichero de entrada (Apilado)'

ERROR: Fin de fichero mientras se leía el grupo 3 en el fichero de entrada (Apilado)'

ERROR: Grupo 3 mal definido (Apilado)'

ERROR: Demasiadas capas en el apilado, excediendo la capacidad del programa'

ERROR: Grupo 4 ausente en el fichero de entrada (Cargas)'

ERROR: Fin de fichero mientras se leía el grupo 4 en el fichero de entrada (Cargas)'

ERROR: Grupo 4 mal definido (Cargas)'

ERROR: Demasiados casos de carga, excediendo la capacidad del programa'

ERROR: División por cero en matriz QQ ($Ex - Ey \cdot V_{xy}^2$)'

ERROR: Valor nulo de A11'

ERROR: Valor nulo de A22'

ERROR: Valor nulo de H'

ERROR: Valor nulo de EX'

ERROR: Valor nulo de EY'

ERROR: Valor nulo de GXY'

ERROR: No se puede abrir el fichero de salida'

6 Windows application

An application has been developed in Visual Basic .NET for Windows environment. The original routines of the program for calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please contact with the system manager.

The application can be executed by "double click" in the file **Eh.exe**, located in the PC's server, or by means a **link** in the "desktop" or in the "start menu".

The input data must be typed in the application form or retrieved from an input file. The form, after loading the data corresponding to the example 1 (see section 7), has the following appearance:

The screenshot shows a Windows application window titled "Eh - Ejemplo-1_1.dat". The window contains several input fields and a table. On the left, there are icons for file operations. The main area has tabs for "Data", "Graph", and "Results". The "Data" tab is selected, showing "Input data". Under "Input data", "Option 1: Laminata" is selected. A table lists material properties for 4 layers. Below the table, there are radio buttons for "Option 2: Membrane stiffness matrix" and "Option 3: Laminata average properties". At the bottom, there are input fields for "Loads" (Fx, Fy, Fxy) and a set of navigation buttons.

#	E1	E2	G12	μ_{12}	t	Theta	et	ec
1	157500	8500	2600	0.3	0.125	0	0.013333	-0.006667
2	157500	8500	2600	0.3	0.125	90	0.013333	-0.006667
3	157500	8500	2600	0.3	0.125	45	0.013333	-0.006667
4	157500	8500	2600	0.3	0.125	-45	0.013333	-0.006667

Option 1: Laminata

Option 2: Membrane stiffness matrix

A11 = A12 = A22 = A33 = Thickness =

Option 3: Laminata average properties

Ex = Ey = Gxy = μ_{xy} =

Loads:

Fx = 100 Fy = -50 Fxy = -100

Figure 2: Input data form.

The analysis will be performed using option 1 for input data, that is, defining a laminated of 32 plies, using four different materials.

The program calculates the results for all the included load cases at the moment it has all the required data. In example 1, there are four load cases and the user can access to the results by clicking on the taps "Graph" and "Results" (See Figures 2 and 3 below).

WARNING

eRef is an “arbitrary” value for plotting purposes. Use graph just for a qualitative understanding of the stress distribution.

Note to be included in future revisions of user's manual (G.B.L. 18/2/2016)

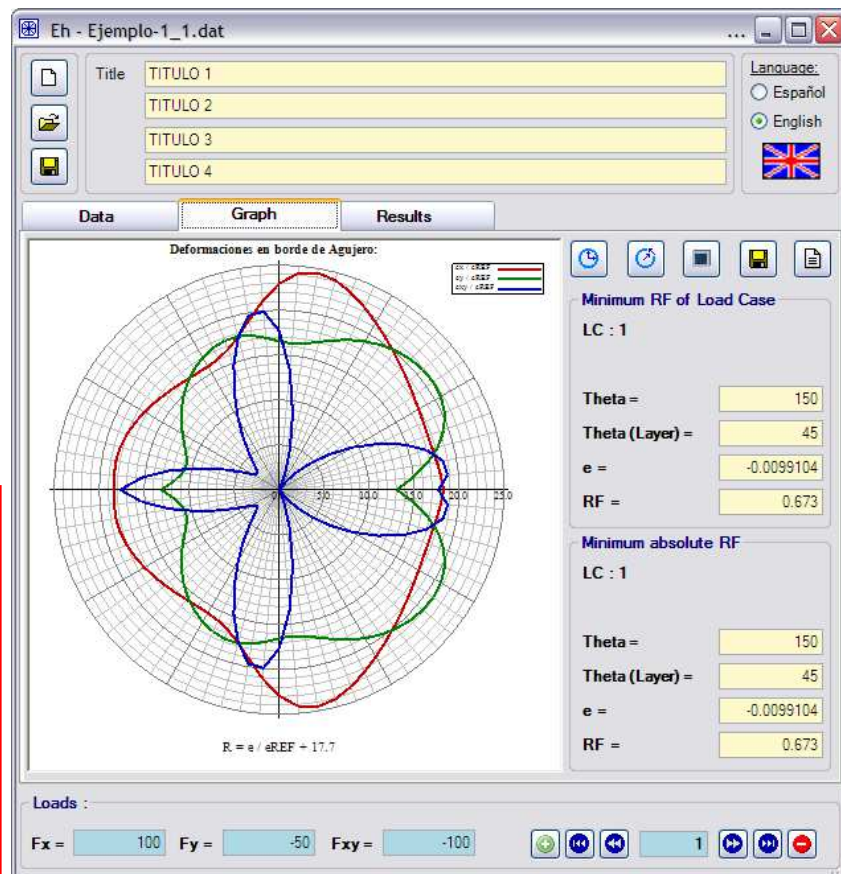


Figure 3: Results form: Graph.

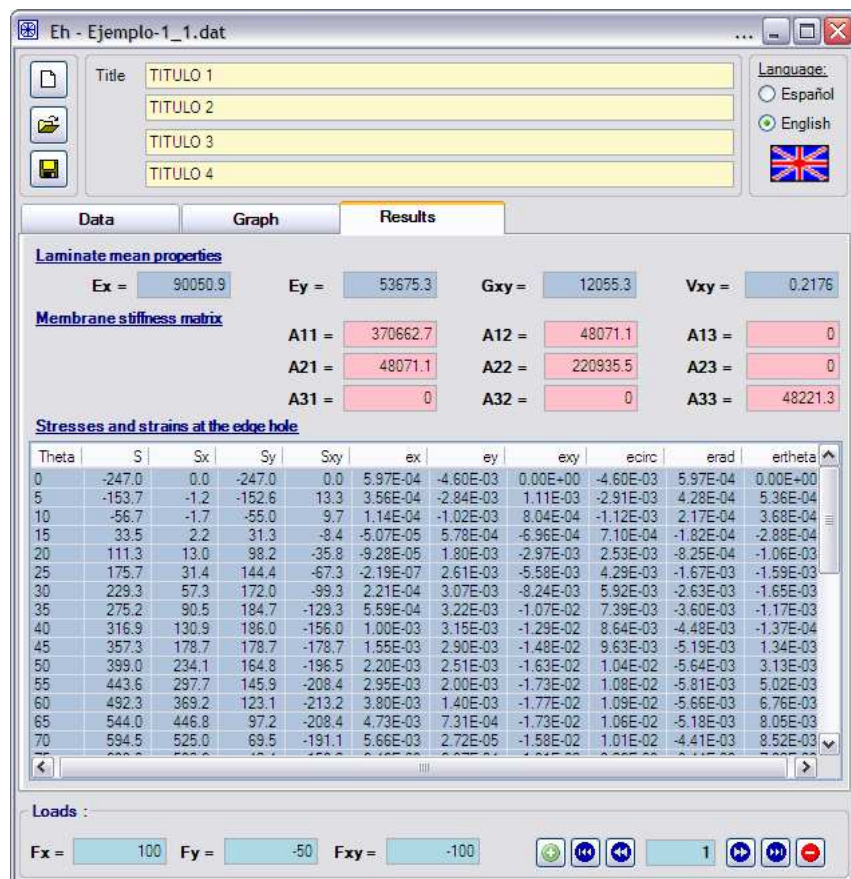


Figure 4: Results form: Results.

7 Example

Three examples of how to use the program are given below. All them correspond to the same situation, but solved in three different ways.

7.1 Formulation of the sample problem

The problem consists of a drilled hole in a plate of composite material. The material of the layers has the following properties:

E_x	175500 MPa
E_y	8500 MPa
G_{xy}	2600 MPa
ν_{xy}	0.30
t	0.125 mm
ϵ_t	0.013333 $\mu\epsilon$
ϵ_c	-0.006667 $\mu\epsilon$

The laminate plate is composed of 32 plies distributed in a symmetric lay-out:

(0/0/0/0/90/90/45/-45/0/0/0/0/90/90/45/-45)_s

Four load cases will be applied over the plate (unit are MPa):

Load Case =====	F_x =====	F_y =====	F_z =====
1	100	-50	-100
2	100	0	0
3	0	-50	0
4	0	0	-100

7.2 Example

The example shows how to solve the problem using the options input data.

General input data file: Option 1

```

1 Example 1: Program EH
  Stresses/Strains Around a Hole in an infinite orthotropic plate
  Input 1: 4 Materials, 32 Layers
  4 Load Cases
2 MATERIAL DATA
  1 Ply Properties
    4 1. 157500.0 8500.0 2600.0 0.300 0.125 0.0 0.013333 -0.006667
      2. 157500.0 8500.0 2600.0 0.300 0.125 90.0 0.013333 -0.006667
      3. 157500.0 8500.0 2600.0 0.300 0.125 45.0 0.013333 -0.006667
      4. 157500.0 8500.0 2600.0 0.300 0.125 -45.0 0.013333 -0.006667

3 LAMINATE
  32 1. 1. 1. 1. 2. 2. 3. 4. 1. 1. 1. 1. 2. 2. 3. 4.
     4. 3. 2. 2. 1. 1. 1. 1. 4. 3. 2. 2. 1. 1. 1. 1.

4 LOADS
  4 100.0 -50.0 -100.0
    100.0 0.0 0.0
      0.0 -50.0 0.0
      0.0 0.0 -100.0

```

General input data file: Option 2

```

1 Example 2: Program EH
  Stresses/Strains Around a Hole in an infinite orthotropic plate
  Input 2: Stiffnes matrix
  4 Load Cases
2 MATERIAL DATA
  2 Stiffness Matrix
    370662.7 48071.1 220935.5 48221.3 4.000

4 LOADS
  4 100.0 -50.0 -100.0
    100.0 0.0 0.0
      0.0 -50.0 0.0
      0.0 0.0 -100.0

```

General input data file: Option 3

```

1 Example 3: Program EH
  Stresses/Strains Around a Hole in an infinite orthotropic plate
  Input 3: Mean properties of the plate
  4 Load Cases
2 MATERIAL DATA
  3 Laminate Mean Properties
    90050.90 53675.30 12055.30 0.22

4 LOADS
  4 100.0 -50.0 -100.0
    100.0 0.0 0.0
      0.0 -50.0 0.0
      0.0 0.0 -100.0

```

Output results file:PROGRAM EH, VERSION 3.0
=====

Example 1: Program EH
Stresses/Strains Around a Hole in an infinite orthotropic plate
Input 1: 4 Materials, 32 Layers
4 Load Cases

PLATE CONSTRUCTION

LAY NUM	LAYER THICKNESS	DIST. FROM NEUTRAL PLN	E11	E22	G12	μ12	LAYER ANGLE DEGREES	ALLOWABLE TENSION	ALLOWABLE COMPRESSION
1	1.2500E-01	1.9375E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
2	1.2500E-01	1.8125E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
3	1.2500E-01	1.6875E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
4	1.2500E-01	1.5625E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
5	1.2500E-01	1.4375E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	9.0000E+01	1.3333E-02	-6.6670E-03
6	1.2500E-01	1.3125E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	9.0000E+01	1.3333E-02	-6.6670E-03
7	1.2500E-01	1.1875E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	4.5000E+01	1.3333E-02	-6.6670E-03
8	1.2500E-01	1.0625E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	-4.5000E+01	1.3333E-02	-6.6670E-03
9	1.2500E-01	9.3750E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
10	1.2500E-01	8.1250E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
11	1.2500E-01	6.8750E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
12	1.2500E-01	5.6250E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
13	1.2500E-01	4.3750E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	9.0000E+01	1.3333E-02	-6.6670E-03
14	1.2500E-01	3.1250E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	9.0000E+01	1.3333E-02	-6.6670E-03
15	1.2500E-01	1.8750E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	4.5000E+01	1.3333E-02	-6.6670E-03
16	1.2500E-01	6.2500E-02	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	-4.5000E+01	1.3333E-02	-6.6670E-03
17	1.2500E-01	-6.2500E-02	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	-4.5000E+01	1.3333E-02	-6.6670E-03
18	1.2500E-01	-1.8750E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	4.5000E+01	1.3333E-02	-6.6670E-03
19	1.2500E-01	-3.1250E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	9.0000E+01	1.3333E-02	-6.6670E-03
20	1.2500E-01	-4.3750E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	9.0000E+01	1.3333E-02	-6.6670E-03
21	1.2500E-01	-5.6250E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
22	1.2500E-01	-6.8750E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
23	1.2500E-01	-8.1250E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
24	1.2500E-01	-9.3750E-01	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
25	1.2500E-01	-1.0625E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	-4.5000E+01	1.3333E-02	-6.6670E-03
26	1.2500E-01	-1.1875E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	4.5000E+01	1.3333E-02	-6.6670E-03
27	1.2500E-01	-1.3125E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	9.0000E+01	1.3333E-02	-6.6670E-03
28	1.2500E-01	-1.4375E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	9.0000E+01	1.3333E-02	-6.6670E-03
29	1.2500E-01	-1.5625E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
30	1.2500E-01	-1.6875E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
31	1.2500E-01	-1.8125E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03
32	1.2500E-01	-1.9375E+00	1.5750E+05	8.5000E+03	2.6000E+03	3.0000E-01	0.0000E+00	1.3333E-02	-6.6670E-03

A IN PLANE STIFFNESS

3.7066E+05 4.8071E+04 0.0000E+00
4.8071E+04 2.2094E+05 0.0000E+00
0.0000E+00 0.0000E+00 4.8221E+04

ESPESOR TOTAL DEL LAMINADO = 4.0000

PROPIEDADES MEDIAS DE LAMINADO

EX EY EXY VXY
9.0051E+04 5.3675E+04 1.2055E+04 2.1758E-01

LOAD CASE : 1 X-STRESS Y-STRESS XY-STRESS
1.0000E+02 -5.0000E+01 -1.0000E+02

THT	SIGMA	XSTRESS	YSTRESS	XYSTRESS	XSTRAIN	YSTRAIN	XYSTRAIN	ECIRCUM	ERAD	ERTHETA
0	-2.4697E+02	0.0000E+00	-2.4697E+02	0.0000E+00	5.9672E-04	-4.6011E-03	0.0000E+00	-4.6011E-03	5.9672E-04	0.0000E+00
5	-1.5374E+02	-1.1678E+00	-1.5257E+02	1.3348E+01	3.5567E-04	-2.8396E-03	1.1072E-03	-2.9115E-03	4.2753E-04	5.3556E-04
10	-5.6673E+01	-1.7089E+00	-5.4964E+01	9.6916E+00	1.1383E-04	-1.0199E-03	8.0393E-04	-1.1232E-03	2.1712E-04	3.6770E-04
15	3.3546E+01	2.2472E+00	3.1299E+01	-8.3865E+00	-5.0670E-05	5.7769E-04	-6.9567E-04	7.0951E-04	-1.8249E-04	-2.8829E-04
20	1.1126E+02	1.3015E+01	9.8243E+01	-3.5757E+01	-9.2848E-05	1.7989E-03	-2.9661E-03	2.5309E-03	-8.2485E-04	-1.0562E-03
25	1.7575E+02	3.1390E+01	1.4436E+02	-6.7316E+01	-2.1945E-07	2.6137E-03	-5.5839E-03	4.2856E-03	-1.6721E-03	-1.5869E-03
30	2.2928E+02	5.7321E+01	1.7196E+02	-9.9282E+01	2.2104E-04	3.0652E-03	-8.2355E-03	5.9203E-03	-2.6340E-03	-1.6546E-03
35	2.7521E+02	9.0541E+01	1.8467E+02	-1.2931E+02	5.5925E-04	3.2217E-03	-1.0726E-02	7.3854E-03	-3.6044E-03	-1.1666E-03
40	3.1690E+02	1.3094E+02	1.8597E+02	-1.5604E+02	1.0047E-03	3.1483E-03	-1.2944E-02	8.6363E-03	-4.4833E-03	-1.3668E-04
45	3.5734E+02	1.7867E+02	1.7867E+02	-1.7867E+02	1.5524E-03	2.8970E-03	-1.4821E-02	9.6350E-03	-5.1856E-03	1.3446E-03
50	3.9896E+02	2.3412E+02	1.6484E+02	-1.9645E+02	2.2016E-03	2.5054E-03	-1.6296E-02	1.0351E-02	-5.6442E-03	3.1289E-03
55	4.4363E+02	2.9768E+02	1.4595E+02	-2.0844E+02	2.9530E-03	1.9999E-03	-1.7290E-02	1.0763E-02	-5.8102E-03	5.0179E-03
60	4.9226E+02	3.6919E+02	1.2306E+02	-2.1315E+02	3.8025E-03	1.4007E-03	-1.7681E-02	1.0858E-02	-5.6551E-03	6.7607E-03
65	5.4401E+02	4.4684E+02	9.7163E+01	-2.0837E+02	4.7274E-03	7.3054E-04	-1.7284E-02	1.0634E-02	-5.1758E-03	8.0483E-03
70	5.9450E+02	5.2496E+02	6.9543E+01	-1.9107E+02	5.6615E-03	2.7234E-05	-1.5849E-02	1.0096E-02	-4.4075E-03	8.5196E-03
75	6.3298E+02	5.9058E+02	4.2402E+01	-1.5825E+02	6.4558E-03	-6.3699E-04	-1.3127E-02	9.2624E-03	-3.4435E-03	7.8216E-03
80	6.3995E+02	6.2065E+02	1.9297E+01	-1.0944E+02	6.8456E-03	-1.1401E-03	-9.0779E-03	8.1572E-03	-2.4517E-03	5.7992E-03
85	5.9119E+02	5.8670E+02	4.4907E+00	-5.1329E+01	6.5043E-03	-1.3339E-03	-4.2578E-03	6.8145E-03	-1.6440E-03	2.8320E-03
90	4.7501E+02	4.7501E+02	1.2608E-11	7.7388E-05	5.2749E-03	-1.1477E-03	6.4194E-09	5.2749E-03	-1.1477E-03	-4.3267E-09
95	3.1104E+02	3.0868E+02	2.3627E+00	2.7006E+01	3.4221E-03	-7.0180E-04	2.2401E-03	3.5853E-03	-8.6497E-04	-1.4900E-03
100	1.4097E+02	1.3672E+02	4.2508E+00	2.4108E+01	1.5080E-03	-2.5115E-04	1.9998E-03	1.7969E-03	-5.4008E-04	-1.2775E-03
105	-2.4430E+00	-2.2794E+00	-1.6365E-01	-6.1076E-01	-2.4917E-05	2.4585E-06	-5.0663E-05	-3.5749E-05	1.3290E-05	3.0188E-05
110	-1.0935E+02	-9.6560E+01	-1.2792E+01	-3.5145E+01	-1.0414E-03	-5.0099E-06	-2.9153E-03	-1.8571E-03	8.1072E-04	1.5671E-03
115	-1.8477E+02	-1.5177E+02	-3.3002E+01	-7.0773E+01	-1.6057E-03	-2.4813E-04	-5.8707E-03	-3.6118E-03	1.7580E-03	2.7337E-03
120	-2.3785E+02	-1.7839E+02	-5.9463E+01	-1.0299E+02	-1.8373E-03	-6.7680E-04	-8.5433E-03	-5.2465E-03	2.7324E-03	3.2666E-03
125	-2.7664E+02	-1.8562E+02	-9.1010E+01	-1.2998E+02	-1.8414E-03	-1.2471E-03	-1.0782E-02	-6.7116E-03	3.6231E-03	3.1209E-03
130	-3.0690E+02	-1.8009E+02	-1.2680E+02	-1.5112E+02	-1.6935E-03	-1.9272E-03	-1.2535E-02	-7.9625E-03	4.3417E-03	2.4069E-03
135	-3.3235E+02	-1.6617E+02	-1.6617E+02	-1.6617E+02	-1.4438E-03	-2.6944E-03	-1.3784E-02	-8.9612E-03	4.8230E-03	1.2506E-03

140	-3.5511E+02	-1.4672E+02	-2.0839E+02	-1.7486E+02	-1.1258E-03	-3.5278E-03	-1.4504E-02	-9.6774E-03	5.0238E-03	-1.5317E-04
145	-3.7597E+02	-1.2369E+02	-2.5228E+02	-1.7665E+02	-7.6401E-04	-4.4013E-03	-1.4653E-02	-1.0089E-02	4.9241E-03	-1.5938E-03
150	-3.9443E+02	-9.8607E+01	-2.9582E+02	-1.7079E+02	-3.8025E-04	-5.2731E-03	-1.4167E-02	-1.0185E-02	4.5312E-03	-2.8464E-03
155	-4.0846E+02	-7.2953E+01	-3.3550E+02	-1.5645E+02	-5.1143E-07	-6.0743E-03	-1.2977E-02	-9.9600E-03	3.8862E-03	-3.6881E-03
160	-4.1422E+02	-4.8454E+01	-3.6576E+02	-1.3313E+02	3.4568E-04	-6.6973E-03	-1.1043E-02	-9.4226E-03	3.0709E-03	-3.9323E-03
165	-4.0607E+02	-2.7202E+01	-3.7887E+02	-1.0152E+02	6.1336E-04	-6.9929E-03	-8.4210E-03	-8.5886E-03	2.2091E-03	-3.4897E-03
170	-3.7760E+02	-1.1386E+01	-3.6621E+02	-6.4573E+01	7.5840E-04	-6.7952E-03	-5.3564E-03	-7.4835E-03	1.4466E-03	-2.4499E-03
175	-3.2425E+02	-2.4630E+00	-3.2179E+02	-2.8153E+01	7.5015E-04	-5.9891E-03	-2.3353E-03	-6.1407E-03	9.0171E-04	-1.1295E-03
180	-2.4697E+02	-2.6221E-11	-2.4697E+02	8.0472E-05	5.9672E-04	-4.6011E-03	6.6752E-09	-4.6011E-03	5.9672E-04	3.2879E-09

STRAINS IN PLIES

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THT	ANGULO CAPA 0.	ANGULO CAPA 90.	ANGULO CAPA 45.	ANGULO CAPA -45.	RF 1	RF 2	RF 3	RF 4	MIN RF
===	=====	=====	=====	=====	=====	=====	=====	=====	=====
0	5.9672E-04	-4.6011E-03	-2.0022E-03	-2.0022E-03	22.34	1.45	3.33	3.33	1.45
5	3.5567E-04	-2.8396E-03	-6.8836E-04	-1.7956E-03	37.49	2.35	9.69	3.71	2.35
10	1.1383E-04	-1.0199E-03	-5.1062E-05	-8.5499E-04	117.13	6.54	130.57	7.80	6.54
15	-5.0670E-05	5.7769E-04	-8.4326E-05	6.1134E-04	131.58	23.08	79.06	21.81	21.81
20	-9.2848E-05	1.7989E-03	-6.3004E-04	2.3361E-03	71.81	7.41	10.58	5.71	5.71
25	-2.1945E-07	2.6137E-03	-1.4852E-03	4.0987E-03	1000.00	5.10	4.49	3.25	3.25
30	2.2104E-04	3.0652E-03	-2.4746E-03	5.7609E-03	60.32	4.35	2.69	2.31	2.31
35	5.5925E-04	3.2217E-03	-3.4725E-03	7.2535E-03	23.84	4.14	1.92	1.84	1.84
40	1.0047E-03	3.1483E-03	-4.3955E-03	8.5485E-03	13.27	4.24	1.52	1.56	1.52
45	1.5524E-03	2.8970E-03	-5.1856E-03	9.6350E-03	8.59	4.60	1.29	1.38	1.29
50	2.2016E-03	2.5054E-03	-5.7944E-03	1.0501E-02	6.06	5.32	1.15	1.27	1.15
55	2.9530E-03	1.9999E-03	-6.1686E-03	1.1122E-02	4.51	6.67	1.08	1.20	1.08
60	3.8025E-03	1.4007E-03	-6.2391E-03	1.1442E-02	3.51	9.52	1.07	1.17	1.07
65	4.7274E-03	7.3054E-04	-5.9131E-03	1.1371E-02	2.82	18.25	1.13	1.17	1.13
70	5.6615E-03	2.7234E-05	-5.0803E-03	1.0769E-02	2.36	489.57	1.31	1.24	1.24
75	6.4558E-03	-6.3699E-04	-3.6539E-03	9.4727E-03	2.07	10.47	1.82	1.41	1.41
80	6.8456E-03	-1.1401E-03	-1.6862E-03	7.3917E-03	1.95	5.85	3.95	1.80	1.80
85	6.5043E-03	-1.3339E-03	4.5631E-04	4.7141E-03	2.05	5.00	29.22	2.83	2.05
90	5.2749E-03	-1.1477E-03	2.0636E-03	2.0636E-03	2.53	5.81	6.46	6.46	2.53
95	3.4221E-03	-7.0180E-04	2.4802E-03	2.4007E-04	3.90	9.50	5.38	55.54	3.90
100	1.5080E-03	-2.5115E-04	1.6283E-03	-3.7145E-04	8.84	26.55	8.19	17.95	8.19
105	-2.4917E-05	2.4585E-06	-3.6561E-05	1.4102E-05	267.57	1000.00	182.35	945.44	182.35
110	-1.0414E-03	-5.0099E-06	-1.9809E-03	3.4466E-04	6.40	1000.00	3.37	14.27	3.37
115	-1.6057E-03	-2.4813E-04	-3.8622E-03	2.0084E-03	4.15	26.87	1.73	6.64	1.73
120	-1.8373E-03	-6.7680E-04	-5.5287E-03	3.0146E-03	3.63	9.85	1.21	4.42	1.21
125	-1.8414E-03	-1.2471E-03	-6.9351E-03	3.8466E-03	3.62	5.35	0.96	3.47	0.96
130	-1.6935E-03	-1.9272E-03	-8.0780E-03	4.4572E-03	3.94	3.46	0.83	2.99	0.83
135	-1.4438E-03	-2.6944E-03	-8.9612E-03	4.8230E-03	4.62	2.47	0.74	2.76	0.74
140	-1.1258E-03	-3.5278E-03	-9.5791E-03	4.9254E-03	5.92	1.89	0.70	2.71	0.70
145	-7.6401E-04	-4.4013E-03	-9.9092E-03	4.7439E-03	8.73	1.51	0.67	2.81	0.67
150	-3.8025E-04	-5.2731E-03	-9.9104E-03	4.2570E-03	17.53	1.26	0.67	3.13	0.67
155	5.1143E-07	-6.0743E-03	-9.5256E-03	3.4518E-03	1000.00	1.10	0.70	3.86	0.70
160	3.4568E-04	-6.6973E-03	-8.6973E-03	2.3457E-03	38.57	1.00	0.77	5.68	0.77
165	6.1336E-04	-6.9929E-03	-7.4003E-03	1.0208E-03	21.74	0.95	0.90	13.06	0.90
170	7.5840E-04	-6.7952E-03	-5.6966E-03	-3.4022E-04	17.58	0.98	1.17	19.60	0.98
175	7.5015E-04	-5.9891E-03	-3.7871E-03	-1.4518E-03	17.77	1.11	1.76	4.59	1.11
180	5.9672E-04	-4.6011E-03	-2.0022E-03	-2.0022E-03	22.34	1.45	3.33	3.33	1.45

MINIMUM RESERVE FACTOR = 0.67 THETA = 45.

LOAD CASE : 2 X-STRESS Y-STRESS XY-STRESS
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THT	SIGMA	XSTRESS	YSTRESS	XYSTRESS	XSTRAIN	YSTRAIN	XYSTRAIN	ECIRCUM	ERAD	ERTHETA
===	=====	=====	=====	=====	=====	=====	=====	=====	=====	=====
0	-7.7205E+01	0.0000E+00	-7.7205E+01	0.0000E+00	1.8654E-04	-1.4384E-03	0.0000E+00	-1.4384E-03	1.8654E-04	0.0000E+00
5	-7.3547E+01	-5.5867E-01	-7.2988E+01	6.3856E+00	1.7015E-04	-1.3585E-03	5.2969E-04	-1.3928E-03	2.0453E-04	2.5621E-04
10	-6.3457E+01	-1.9135E+00	-6.1543E+01	1.0852E+01	1.2745E-04	-1.1420E-03	9.0016E-04	-1.2576E-03	2.4311E-04	4.1171E-04
15	-4.9022E+01	-3.2839E+00	-4.5738E+01	1.2256E+01	7.4046E-05	-8.4420E-04	1.0166E-03	-1.0368E-03	2.6669E-04	4.2129E-04
20	-3.2407E+01	-3.7909E+00	-2.8616E+01	1.0415E+01	2.7045E-05	-5.2398E-04	8.6397E-04	-7.3719E-04	2.4026E-04	3.0765E-04
25	-1.5083E+01	-2.6939E+00	-1.2389E+01	5.7771E+00	1.8833E-08	-2.2430E-04	4.7921E-04	-3.6779E-04	1.4350E-04	1.3619E-04
30	2.3295E+00	5.8239E-01	1.7472E+00	-1.0087E+00	2.2458E-06	3.1143E-05	-8.3674E-05	6.0151E-05	-2.6762E-05	-1.6811E-05
35	1.9885E+01	6.5419E+00	1.3343E+01	-9.3429E+00	4.0408E-05	2.3278E-04	-7.7500E-04	5.3362E-04	-2.6043E-04	-8.4294E-05
40	3.8097E+01	1.5741E+01	2.2357E+01	-1.8759E+01	1.2078E-04	3.7848E-04	-1.5561E-03	1.0382E-03	-5.3897E-04	-1.6432E-05
45	5.7807E+01	2.8903E+01	2.8903E+01	-2.8903E+01	2.5113E-04	4.6865E-04	-2.3976E-03	1.5587E-03	-8.3889E-04	2.1752E-04
50	8.0134E+01	4.7024E+01	3.3109E+01	-3.9458E+01	4.4220E-04	5.0322E-04	-3.2731E-03	2.0791E-03	-1.1337E-03	6.2846E-04
55	1.0649E+02	7.1459E+01	3.5035E+01	-5.0036E+01	7.0888E-04	4.8007E-04	-4.1505E-03	2.5837E-03	-1.3948E-03	1.2045E-03
60	1.3860E+02	1.0395E+02	3.4649E+01	-6.0014E+01	1.0706E-03	3.9438E-04	-4.9782E-03	3.0572E-03	-1.5922E-03	1.9035E-03
65	1.7829E+02	1.4645E+02	3.1844E+01	-6.8290E+01	1.5494E-03	2.3943E-04	-5.6648E-03	3.4851E-03	-1.6963E-03	2.6378E-03
70	2.2697E+02	2.0042E+02	2.6550E+01	-7.2945E+01	2.1614E-03	1.0397E-05	-6.0509E-03	3.8545E-03	-1.6827E-03	3.2526E-03
75	2.8389E+02	2.6488E+02	1.9017E+01	-7.0973E+01	2.8954E-03	-2.8569E-04	-5.8873E-03	4.1542E-03	-1.5444E-03	3.5080E-03
80	3.4322E+02	3.3287E+02	1.0349E+01	-5.8694E+01	3.6715E-03	-6.1147E-04	-4.8688E-03	4.3750E-03	-1.3149E-03	3.1103E-03
85	3.9128E+02	3.8831E+02	2.9722E+00	-3.3972E+01	4.3049E-03	-8.8285E-04	-2.8180E-03	4.1502E-03	-1.0881E-03	1.8744E-03
90	4.1024E+02	4.1024E+02	1.0889E-11	6.6837E-05	4.5557E-03	-9.9123E-04	5.5442E-09	4.5557E-03	-9.9123E-04	-3.7368E-09
95	3.9128E+02	3.8831E+02	2.9722E+00	3.3972E+01	4.3049E-03	-8.8285E-04	2.8180E-03	4.1502E-03	-1.0881E-03	-1.8744E-03
100	3.4322E+02	3.3287E+02	1.0349E+01	5.8695E+01	3.6715E-03	-6.1147E-04	4.8688E-03	4.3750E-03	-1.3149E-03	-3.1103E-03
105	2.8389E+02	2.6488E+02	1.9017E+01	7.0973E+01	2.8954E-03	-2.8569E-04	5.8873E-03	4.1542E-03	-1.5444E-03	-3.5080E-03
110	2.2697E+02	2.0042E+02	2.6550E+01	7.2945E+01	2.1614E-03	1.0398E-05	6.0509E-03	3.8545E-03	-1.6827E-03	-3.2526E-03
115	1.7829E+02	1.4645E+02	3.1844E+01	6.8290E+01	1.5494E-03	2.3943E-04	5.6648E-03	3.4851E-03	-1.6963E-03	-2.6378E-03
120	1.3860E+02	1.0395E+02	3.4649E+01	6.0014E+01	1.0706E-03	3.9438E-04	4.9782E-03	3.0572E-03	-1.5922E-03	1.9035E-03
125	1.0649E+02	7.1458E+01	3.5035E+01	5.0036E+01	7.0888E-04	4.8007E-04	4.1505E-03	2.5837E-03	-1.3948E-03	1.2045E-03
130	8.0134E+01	4.7024E+01	3.3109E+01	3.9458E+01	4.4220E-04	5.0322E-04	-3.2731E-03	2.0791E-03	-1.1337E-03	6.2846E-04
135	5.7807E+01	2.8903E+01	2.8903E+01	-2.8903E+01	2.5113E-04	4.6865E-04	-2.3976E-03	1.5587E-03	-8.3889E-04	-2.1752E-04
140	3.8097E+01	1.5741E+01	2.2356E+01	1.8759E+01	1.2078E-04	3.7848E-04	1.5561E-03	1.0382E-03	-5.3897E-04	1.6432E-05
145	1.9885E+01	6.5419E+00	1.3343E+01	9.3428E+00	4.0408E-05	2.3278E-04	7.7499E-04	5.3362E-04	-2.6043E-04	8.4294E-05
150	2.3295E+00	5.8237E-01	1.7471E+00	1.0087E+00	2.2458E-06	3.1143E-05	8.3672E-05	6.0149E-05	-2.6761E-05	1.6811E-05
155	-1.5083E+01	-2.6939E+00	-1.2389E+01	-5.7771E+00	1.8886E-08	-2.2431E-04	-4.7921E-04	-3.6779E-04	1.4350E-04	-1.3619E-04
160	-3.2407E+01	-3.7909E+00	-2.8616E+01	-1.0415E+01	2.7045E-05	-5.2398E-04	-8.6397E-04	-7.3719E-04	2.4026E-04	-3.0765E-04
165	-4.9022E+01	-3.2839E+00	-4.5738E+01	-1.2256E+01	7.4046E-05	-8.4420E-04	-1.0166E-03	-1.0368E-03	2.6669E-04	-4.2129E-04
170	-6.3457E+01	-1.9135E+00	-6.1543E+01	-1.0852E+01	1.2745E-04	-1.1420E-03	-9.0016E-04	-1.2576E-03	2.4311E-04	-4.1171E-04
175	-7.3547E+01	-5.5866E-01	-7.2988E+01	-6.3856E+00	1.7015E-04	-1.3585E-03	-5.2969E-04	-1.3928E-03	2.0453E-04	-2.5620E-04
180	-7.7205E+01	-8.1970E-12	-7.7205E+01	2.5156E-05	1.8654E-04	-1.4384E-03	2.0867E-09	-1.4384E-03	1.8654E-04	1.0278E-09

THT	ANGULO CAPA 0.	ANGULO CAPA 90.	ANGULO CAPA 45.	ANGULO CAPA -45.	RF 1	RF 2	RF 3	RF 4	MIN RF
===	=====	=====	=====	=====	=====	=====	=====	=====	=====
0	1.8654E-04	-1.4384E-03	-6.2591E-04	-6.2591E-04	71.47	4.64	10.65	10.65	4.64
5	1.7015E-04	-1.3585E-03	-3.2931E-04	-8.5900E-04	78.36	4.91	20.25	7.76	4.91
10	1.2745E-04	-1.1420E-03	-5.7174E-05	-9.5734E-04	104.61	5.84	116.61	6.96	5.84
15	7.4046E-05	-8.4420E-04	1.2323E-04	-8.9338E-04	180.06	7.90	108.20	7.46	7.46
20	2.7045E-05	-5.2398E-04	1.8352E-04	-6.8045E-04	493.00	12.72	72.65	9.80	9.80
25	1.8833E-08	-2.2430E-04	1.2746E-04	-3.5175E-04	1000.00	29.72	104.60	18.95	18.95
30	2.2458E-06	3.1143E-05	-2.5143E-05	5.8532E-05	1000.00	428.12	265.17	227.79	227.79
35	4.0408E-05	2.3278E-04	-2.5090E-04	5.2409E-04	329.96	57.28	26.57	25.44	25.44
40	1.2078E-04	3.7848E-04	-5.2842E-04	1.0277E-03	110.39	35.23	12.62	12.97	12.62
45	2.5113E-04	4.6865E-04	-8.3889E-04	1.5587E-03	53.09	28.45	7.95	8.55	7.95
50	4.4220E-04	5.0322E-04	-1.1638E-03	2.1093E-03	30.15	26.50	5.73	6.32	5.73
55	7.0888E-04	4.8007E-04	-1.4808E-03	2.6697E-03	18.81	27.77	4.50	4.99	4.50
60	1.0706E-03	3.9438E-04	-1.7566E-03	3.2216E-03	12.45	33.81	3.80	4.14	3.80
65	1.5494E-03	2.3943E-04	-1.9380E-03	3.7268E-03	8.61	55.69	3.44	3.58	3.44
70	2.1614E-03	1.0397E-05	-1.9395E-03	4.1114E-03	6.17	1000.00	3.44	3.24	3.24
75	2.8954E-03	-2.8569E-04	-1.6388E-03	4.2485E-03	4.60	23.34	4.07	3.14	3.14
80	3.6715E-03	-6.1147E-04	-9.0436E-04	3.9644E-03	3.63	10.90	7.37	3.36	3.36
85	4.3049E-03	-8.8285E-04	3.0201E-04	3.1200E-03	3.10	7.55	44.15	4.27	3.10
90	4.5557E-03	-9.9123E-04	1.7822E-03	1.7822E-03	2.93	6.73	7.48	7.48	2.93
95	4.3049E-03	-8.8285E-04	3.1200E-03	3.0200E-04	3.10	7.55	4.27	44.15	3.10
100	3.6715E-03	-6.1147E-04	3.9644E-03	-9.0437E-04	3.63	10.90	3.36	7.37	3.36
105	2.8954E-03	-2.8569E-04	4.2485E-03	-1.6388E-03	4.60	23.34	3.14	4.07	3.14
110	2.1614E-03	1.0398E-05	4.1114E-03	-1.9395E-03	6.17	1000.00	3.24	3.44	3.24
115	1.5494E-03	2.3943E-04	3.7268E-03	-1.9380E-03	8.61	55.69	3.58	3.44	3.44
120	1.0706E-03	3.9438E-04	3.2216E-03	-1.7566E-03	12.45	33.81	4.14	3.80	3.80
125	7.0888E-04	4.8007E-04	2.6697E-03	-1.4808E-03	18.81	27.77	4.99	4.50	4.50
130	4.4220E-04	5.0322E-04	2.1093E-03	-1.1638E-03	30.15	26.50	6.32	5.73	5.73
135	2.5113E-04	4.6865E-04	1.5587E-03	-8.3889E-04	53.09	28.45	8.55	7.95	7.95
140	1.2078E-04	3.7848E-04	1.0277E-03	-5.2842E-04	110.39	35.23	12.97	12.62	12.62
145	4.0408E-05	2.3278E-04	5.2409E-04	-2.5090E-04	329.96	57.28	25.44	26.57	25.44
150	2.2458E-06	3.1143E-05	5.8530E-05	-2.5142E-05	1000.00	428.13	227.80	265.17	227.80
155	1.8886E-08	-2.2431E-04	-3.5175E-04	1.2746E-04	1000.00	29.72	18.95	104.60	18.95
160	2.7045E-05	-5.2398E-04	-6.8045E-04	1.8352E-04	493.00	12.72	9.80	72.65	9.80
165	7.4046E-05	-8.4420E-04	-8.9338E-04	1.2323E-04	180.06	7.90	7.46	108.20	7.46
170	1.2745E-04	-1.1420E-03	-9.5734E-04	-5.7175E-05	104.61	5.84	6.96	116.61	5.84
175	1.7015E-04	-1.3585E-03	-3.2931E-04	-8.5900E-04	78.36	4.91	7.76	20.25	4.91
180	1.8654E-04	-1.4384E-03	-6.2591E-04	-6.2591E-04	71.47	4.64	10.65	10.65	4.64

MINIMUM RESERVE FACTOR = 2.93 THETA = 0.

LOAD CASE :	3	X-STRESS	Y-STRESS	XY-STRESS
		=====	=====	=====
		0.0000E+00	-5.0000E+01	0.0000E+00

THT	SIGMA	XSTRESS	YSTRESS	XYSTRESS	XSTRAIN	YSTRAIN	XYSTRAIN	ECIRCUM	ERAD	ERTHETA
===	=====	=====	=====	=====	=====	=====	=====	=====	=====	=====
0	-1.6976E+02	0.0000E+00	-1.6976E+02	0.0000E+00	4.1018E-04	-3.1627E-03	0.0000E+00	-3.1627E-03	4.1018E-04	0.0000E+00
5	-1.6545E+02	-1.2568E+00	-1.6419E+02	1.4365E+01	3.8276E-04	-3.0559E-03	1.1916E-03	-3.1333E-03	4.6010E-04	5.7635E-04
10	-1.5368E+02	-4.6340E+00	-1.4904E+02	2.6281E+01	3.0866E-04	-2.7656E-03	2.1800E-03	-3.0457E-03	5.8876E-04	9.9707E-04
15	-1.3724E+02	-9.1934E+00	-1.2805E+02	3.4310E+01	2.0730E-04	-2.3634E-03	2.8461E-03	-2.9027E-03	7.4661E-04	1.1794E-03
20	-1.1907E+02	-1.3929E+01	-1.0514E+02	3.8269E+01	9.9369E-05	-1.9252E-03	3.1745E-03	-2.7086E-03	8.8279E-04	1.1304E-03
25	-1.0127E+02	-1.8087E+01	-8.3182E+01	3.8789E+01	1.2644E-07	-1.5060E-03	3.2175E-03	-2.4694E-03	9.6351E-04	9.1441E-04
30	-8.4903E+01	-2.1226E+01	-6.3677E+01	3.6764E+01	-8.1852E-05	-1.1351E-03	3.0496E-03	-2.1923E-03	9.7536E-04	6.1270E-04
35	-7.0266E+01	-2.3117E+01	-4.7149E+01	3.3014E+01	-1.4279E-04	-8.2256E-04	2.7386E-03	-1.8856E-03	9.2028E-04	2.9786E-04
40	-5.7200E+01	-2.3634E+01	-3.3566E+01	2.8166E+01	-1.8135E-04	-5.6826E-04	2.3364E-03	-1.5588E-03	8.0922E-04	2.4671E-05
45	-4.5313E+01	-2.2656E+01	-2.2656E+01	2.2656E+01	-1.9685E-04	-3.6736E-04	1.8794E-03	-1.2218E-03	6.5757E-04	-1.7050E-04
50	-3.4100E+01	-2.0011E+01	-1.4089E+01	1.6791E+01	-1.8817E-04	-2.1414E-04	1.3928E-03	-8.8474E-04	4.8242E-04	-2.6744E-04
55	-2.2997E+01	-1.5431E+01	-7.5657E+00	1.0805E+01	-1.5308E-04	-1.0367E-04	8.9627E-03	-5.5793E-04	3.0119E-04	-2.6011E-04
60	-1.1393E+01	-8.5445E+00	-2.8482E+00	4.9332E+00	-8.8004E-05	-3.2418E-05	4.0921E-04	-2.5130E-04	1.3088E-04	-1.5647E-04
65	1.3222E+00	1.0861E+00	2.3616E-01	-5.0644E-01	1.1490E-05	1.7756E-06	-4.2010E-05	2.5846E-05	-1.2580E-05	1.9562E-05
70	1.5609E+01	1.3783E+01	1.8259E+00	-5.0166E+00	1.4865E-04	7.1504E-07	-4.1613E-04	2.6508E-04	-1.1572E-04	2.2369E-04
75	3.1377E+01	2.9275E+01	2.1019E+00	-7.8443E+00	3.2002E-04	-3.1576E-05	-6.5069E-04	4.5914E-04	-1.7070E-04	3.8772E-04
80	4.7238E+01	4.5814E+01	1.4244E+00	-8.0782E+00	5.0531E-04	-8.4157E-05	-6.7009E-04	6.0213E-04	-1.8097E-04	4.2807E-04
85	5.9834E+01	5.9380E+01	4.5451E-01	-5.1951E+00	6.5831E-04	-1.3501E-04	-4.3093E-04	6.8969E-04	-1.6639E-04	2.8663E-04
90	6.4763E+01	6.4763E+01	1.7190E-12	1.0551E-05	7.1918E-04	-1.5648E-04	8.7523E-10	7.1918E-04	-1.5648E-04	-5.8991E-10
95	5.9834E+01	5.9380E+01	4.5451E-01	5.1951E+00	6.5830E-04	-1.3500E-04	4.3094E-04	6.8969E-04	-1.6639E-04	-2.8663E-04
100	4.7238E+01	4.5813E+01	1.4244E+00	8.0782E+00	5.0531E-04	-8.4157E-05	6.7009E-04	6.0213E-04	-1.8097E-04	-4.2807E-04
105	3.1377E+01	2.9275E+01	2.1019E+00	7.8443E+00	3.2002E-04	-3.1576E-05	6.5069E-04	4.5914E-04	-1.7070E-04	-3.8772E-04
110	1.5609E+01	1.3783E+01	1.8259E+00	5.0166E+00	1.4865E-04	7.1511E-07	4.1613E-04	2.6508E-04	-1.1572E-04	-2.2369E-04
115	1.3222E+00	1.0860E+00	2.3615E-01	5.0642E-01	1.1490E-05	1.7755E-06	4.2008E-05	2.5845E-05	-1.2580E-05	-1.9561E-05
120	-1.1393E+01	-8.5445E+00	-2.8482E+00	-4.9332E+00	-8.8004E-05	-3.2418E-05	-4.0921E-04	-2.5130E-04	1.3088E-04	1.5647E-04
125	-2.2997E+01	-1.5431E+01	-7.5657E+00	-1.0805E+01	-1.5308E-04	-1.0367E-04	-8.9628E-04	-5.5793E-04	3.0119E-04	2.6011E-04
130	-3.4100E+01	-2.0011E+01	-1.4089E+01	-1.6791E+01	-1.8817E-04	-2.1414E-04	-1.3928E-03	-8.8474E-04	4.8242E-04	2.6744E-04
135	-4.5313E+01	-2.2656E+01	-2.2656E+01	-2.2656E+01	-1.9685E-04	-3.6736E-04	-1.8794E-03	-1.2218E-03	6.5757E-04	1.7050E-04
140	-5.7200E+01	-2.3634E+01	-3.3566E+01	-2.8166E+01	-1.8135E-04	-5.6826E-04	-2.3364E-03	-1.5588E-03	8.0922E-04	-2.4672E-05
145	-7.0266E+01	-2.3117E+01	-4.7149E+01	-3.3014E+01	-1.4279E-04	-8.2257E-04	-2.7386E-03	-1.8856E-03	9.2028E-04	-2.9787E-04
150	-8.4903E+01	-2.1226E+01	-6.3677E+01	-3.6764E+01	-8.1852E-05	-1.1351E-03	3.0496E-03	-2.1923E-03	9.7536E-04	-6.1270E-04
155	-1.0127E+02	-1.8087E+01	-8.3182E+01	-3.8789E+01	1.2680E-07	-1.5060E-03	-3.2175E-03	-2.4694E-03	9.6351E-04	-9.1442E-04
160	-1.1907E+02	-1.3929E+01	-1.0514E+02	-3.8269E+01	9.9370E-05	-1.9252E-03	3.1745E-03	-2.7086E-03	8.8279E-04	-1.1304E-03
165	-1.3724E+02	-9.1934E+00	-1.2805E+02	-3.4310E+01	2.0730E-04	-2.3634E-03	-2.8461E-03	-2.9027E-03	7.4661E-04	-1.1794E-03
170	-1.5368E+02	-4.6340E+00	-1.4904E+02	-2.6281E+01	3.0866E-04	-2.7656E-03	-2.1800E-03	-3.0457E-03	5.8876E-04	-9.9707E-04
175	-1.6545E+02	-1.2567E+00	-1.6419E+02	-1.4365E+01	3.8276E-04	-3.0559E-03	-1.1916E-03	-3.1333E-03	4.6010E-04	-5.7635E-04
180	-1.6976E+02	-1.8024E-11	-1.6976E+02	5.5315E-05	4.1018E-04	-3.1627E-03	4.5885E-09	-3.1627E-03	4.1018E-04	2.2600E-09

STRAINS IN PLIES

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THT	ANGULO CAPA 0.	ANGULO CAPA 90.	ANGULO CAPA 45.	ANGULO CAPA -45.	RF 1	RF 2	RF 3	RF 4	MIN RF
===	=====	=====	=====	=====	=====	=====	=====	=====	=====
0	4.1018E-04	-3.1627E-03	-1.3763E-03	-1.3763E-03	32.51	2.11	4.84	4.84	2.11
5	3.8276E-04	-3.0559E-03	-7.4080E-04	-1.9324E-03	34.83	2.18	9.00	3.45	2.18
10	3.0866E-04	-2.7656E-03	-1.3846E-04	-2.3185E-03	43.20	2.41	48.15	2.88	2.41
15	2.0730E-04	-2.3634E-03	3.4499E-04	-2.5011E-03	64.32	2.82	38.65	2.67	2.67
20	9.9369E-05	-1.9252E-03	6.7430E-04	-2.5002E-03	134.18	3.46	19.77	2.67	2.67
25	1.2644E-07	-1.5060E-03	8.5582E-04	-2.3617E-03	1000.00	4.43	15.58	2.82	2.82
30	-8.1852E-05	-1.1351E-03	9.1635E-04	-2.1333E-03	81.45	5.87	14.55	3.13	3.13
35	-1.4279E-04	-8.2256E-04	8.8661E-04	-1.8520E-03	46.69	8.11	15.04	3.60	3.60
40	-1.8135E-04	-5.6826E-04	7.9338E-04	-1.5430E-03	36.76	11.73	16.81	4.32	4.32

45	-1.9685E-04	-3.6736E-04	6.5757E-04	-1.2218E-03	33.87	18.15	20.28	5.46	5.46
50	-1.8817E-04	-2.1414E-04	4.9526E-04	-8.9757E-04	35.43	31.13	26.92	7.43	7.43
55	-1.5308E-04	-1.0367E-04	3.1976E-04	-5.7651E-04	43.55	64.31	41.70	11.56	11.56
60	-8.8004E-05	-3.2418E-05	1.4439E-04	-2.6482E-04	75.76	205.66	92.34	25.18	25.18
65	1.1490E-05	1.7756E-06	-1.4372E-05	2.7638E-05	1000.00	1000.00	463.88	482.42	463.88
70	1.4865E-04	7.1504E-07	-1.3338E-04	2.8275E-04	89.70	1000.00	49.98	47.16	47.16
75	3.2002E-04	-3.1576E-05	-1.8112E-04	4.6957E-04	41.66	211.14	36.81	28.39	28.39
80	5.0531E-04	-8.4157E-05	-1.2447E-04	5.4562E-04	26.39	79.22	53.56	24.44	24.44
85	6.5831E-04	-1.3501E-04	4.6183E-05	4.7712E-04	20.25	49.38	288.70	27.94	20.25
90	7.1918E-04	-1.5648E-04	2.8135E-04	2.8135E-04	18.54	42.61	47.39	47.39	18.54
95	6.5830E-04	-1.3500E-04	4.7712E-04	4.6182E-05	20.25	49.38	27.94	288.70	20.25
100	5.0531E-04	-8.4157E-05	5.4562E-04	-1.2447E-04	26.39	79.22	24.44	53.56	24.44
105	3.2002E-04	-3.1576E-05	4.6957E-04	-1.8112E-04	41.66	211.14	28.39	36.81	28.39
110	1.4865E-04	7.1511E-07	2.8274E-04	-1.3338E-04	89.70	1000.00	47.16	49.98	47.16
115	1.1490E-05	1.7755E-06	2.7637E-05	-1.4372E-05	1000.00	1000.00	482.44	463.90	463.90
120	-8.8004E-05	-3.2418E-05	-2.6482E-04	1.4440E-04	75.76	205.66	25.18	92.34	25.18
125	-1.5308E-04	-1.0367E-04	-5.7651E-04	3.1976E-04	43.55	64.31	11.56	41.70	11.56
130	-1.8817E-04	-2.1414E-04	-8.9757E-04	4.9526E-04	35.43	31.13	7.43	26.92	7.43
135	-1.9685E-04	-3.6736E-04	-1.2218E-03	6.5757E-04	33.87	18.15	5.46	20.28	5.46
140	-1.8135E-04	-5.6826E-04	-1.5430E-03	7.9338E-04	36.76	11.73	4.32	16.81	4.32
145	-1.4279E-04	-8.2257E-04	-1.8520E-03	8.8661E-04	46.69	8.11	3.60	15.04	3.60
150	-8.1852E-05	-1.1351E-03	-2.1333E-03	9.1635E-04	81.45	5.87	3.13	14.55	3.13
155	1.2680E-07	-1.5060E-03	-2.3617E-03	8.5582E-04	1000.00	4.43	2.82	15.58	2.82
160	9.9370E-05	-1.9252E-03	-2.5002E-03	6.7430E-04	134.18	3.46	2.67	19.77	2.67
165	2.0730E-04	-2.3634E-03	-2.5011E-03	3.4499E-04	64.32	2.82	2.67	38.65	2.67
170	3.0866E-04	-2.7656E-03	-2.3185E-03	-1.3847E-04	43.20	2.41	2.88	48.15	2.41
175	3.8276E-04	-3.0559E-03	-1.9324E-03	-7.4080E-04	34.83	2.18	3.45	9.00	2.18
180	4.1018E-04	-3.1627E-03	-1.3763E-03	-1.3763E-03	32.51	2.11	4.84	4.84	2.11

MINIMUM RESERVE FACTOR = 2.11 THETA = 90.

LOAD CASE : 4 X-STRESS Y-STRESS XY-STRESS
=====

THT	SIGMA	XSTRESS	YSTRESS	XYSTRESS	XSTRAIN	YSTRAIN	XYSTRAIN	ECIRCUM	ERAD	ERTHETA
0	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
5	8.5257E+01	6.4762E-01	8.4609E+01	-7.4023E+00	1.5747E-03	-6.1403E-04	1.6146E-03	-2.3709E-04	-2.9700E-04	-2.9700E-04
10	1.6046E+02	4.8386E+00	1.5562E+02	-2.7441E+01	-3.2229E-04	2.8877E-03	-2.2762E-03	3.1801E-03	-6.1475E-04	-1.0411E-03
15	2.1981E+02	1.4724E+01	2.0509E+02	-5.4952E+01	-3.3201E-04	3.7853E-03	-4.5584E-03	4.6491E-03	-1.1958E-03	-1.8890E-03
20	2.6274E+02	3.0734E+01	2.3200E+02	-8.4442E+01	-2.1926E-04	4.2481E-03	-7.0045E-03	5.9767E-03	-1.9479E-03	-2.4942E-03
25	2.9210E+02	5.2171E+01	2.3993E+02	-1.1188E+02	-3.6473E-07	4.3440E-03	-9.2807E-03	7.1228E-03	-2.7791E-03	-2.6375E-03
30	3.1186E+02	7.7964E+01	2.3389E+02	-1.3504E+02	3.0065E-04	4.1692E-03	-1.1201E-02	8.0524E-03	-3.5826E-03	-2.2505E-03
35	3.2559E+02	1.0712E+02	2.1847E+02	-1.5298E+02	6.6163E-04	3.8115E-03	-1.2690E-02	8.7374E-03	-4.2643E-03	-1.3802E-03
40	3.3601E+02	1.3883E+02	1.9718E+02	-1.6545E+02	1.0653E-03	3.381E-03	-1.3724E-02	9.1569E-03	-4.7535E-03	-1.4492E-04
45	3.4484E+02	1.7242E+02	1.7242E+02	-1.7242E+02	1.4981E-03	2.7957E-03	-1.4302E-02	9.2981E-03	-5.0043E-03	1.2976E-03
50	3.5293E+02	2.0711E+02	1.4582E+02	-1.7378E+02	1.9476E-03	2.2163E-03	-1.4415E-02	9.1569E-03	-4.9930E-03	2.7679E-03
55	3.6013E+02	2.4165E+02	1.1848E+02	-1.6921E+02	2.3972E-03	1.6235E-03	-1.4036E-02	8.7374E-03	-4.7167E-03	4.0734E-03
60	3.6505E+02	2.7379E+02	9.1264E+01	-1.5807E+02	2.8199E-03	1.0388E-03	-1.3112E-02	8.0524E-03	-4.1937E-03	5.0136E-03
65	3.6439E+02	2.9931E+02	6.5082E+01	-1.3957E+02	3.1665E-03	4.8934E-04	-1.1577E-02	7.1228E-03	-3.4669E-03	5.3910E-03
70	3.5193E+02	3.1076E+02	4.1168E+01	-1.1311E+02	3.3515E-03	1.6122E-05	-9.3823E-03	5.9767E-03	-2.6091E-03	5.0434E-03
75	3.1771E+02	2.9643E+02	2.1283E+01	-7.9428E+01	3.2404E-03	-3.1972E-04	-6.5886E-03	4.6491E-03	-1.7284E-03	3.9259E-03
80	2.4949E+02	2.4196E+02	7.5230E+00	-4.2665E+01	2.6688E-03	-4.4447E-04	-3.5391E-03	3.1801E-03	-9.5582E-04	2.2608E-03
85	1.4007E+02	1.3901E+02	1.0640E+00	-1.2162E+01	1.5411E-03	-3.1605E-04	-1.0088E-03	1.6146E-03	-3.8953E-04	6.7101E-04
90	-2.7283E-04	-2.7283E-04	-7.2417E-18	-4.4449E-11	-3.0297E-09	6.5920E-10	-3.6871E-15	-3.0297E-09	6.5920E-10	6.2851E-15
95	-1.4007E+02	-1.3901E+02	-1.0640E+00	-1.2162E+01	-1.5411E-03	3.1605E-04	-1.0088E-03	-1.6146E-03	3.8954E-04	6.7102E-04
100	-2.4949E+02	-2.4196E+02	-7.5230E+00	-4.2665E+01	-2.6688E-03	4.4447E-04	-3.5391E-03	-3.1801E-03	9.5582E-04	2.2609E-03
105	-3.1771E+02	-2.9643E+02	-2.1283E+01	-7.9428E+01	-3.2404E-03	3.1972E-04	-6.5886E-03	-4.6491E-03	1.7284E-03	3.9259E-03
110	-3.5193E+02	-3.1076E+02	-4.1168E+01	-1.1311E+02	-3.3515E-03	-1.6123E-05	-9.3823E-03	-5.9767E-03	2.6091E-03	5.0434E-03
115	-3.6439E+02	-2.9931E+02	-6.5082E+01	-1.3957E+02	-3.1665E-03	-4.8934E-04	-1.1577E-02	-7.1228E-03	3.4669E-03	5.3910E-03
120	-3.6505E+02	-2.7379E+02	-9.1264E+01	-1.5807E+02	-2.8199E-03	-1.0388E-03	-1.3112E-02	-8.0524E-03	4.1938E-03	5.0136E-03
125	-3.6013E+02	-2.4165E+02	-1.1848E+02	-1.6921E+02	-2.3972E-03	-1.6235E-03	-1.4036E-02	-8.7374E-03	4.7167E-03	4.0734E-03
130	-3.5293E+02	-2.0711E+02	-1.4582E+02	-1.7378E+02	-1.9476E-03	-2.2163E-03	-1.4415E-02	-9.1569E-03	4.9930E-03	2.7679E-03
135	-3.4484E+02	-1.7242E+02	-1.7242E+02	-1.7242E+02	-1.4981E-03	-2.7957E-03	-1.4302E-02	-9.2981E-03	5.0043E-03	1.2976E-03
140	-3.3600E+02	-1.3883E+02	-1.9718E+02	-1.6545E+02	-1.0653E-03	-3.381E-03	-1.3724E-02	-9.1569E-03	4.7535E-03	-1.4493E-04
145	-3.2559E+02	-1.0712E+02	-2.1847E+02	-1.5298E+02	-6.6163E-04	-3.8115E-03	-1.2690E-02	-8.7374E-03	4.2643E-03	-1.3802E-03
150	-3.1186E+02	7.7964E+01	2.3389E+02	-1.3504E+02	3.0065E-04	4.1692E-03	-1.1201E-02	-8.0524E-03	3.5826E-03	-2.2505E-03
155	-2.9210E+02	-5.2171E+01	-2.3993E+02	-1.1188E+02	3.6575E-07	-4.3440E-03	-9.2807E-03	-7.1228E-03	2.7791E-03	-2.6375E-03
160	-2.6274E+02	-3.0734E+01	-2.3200E+02	-8.4442E+01	2.1926E-04	-4.2481E-03	-7.0045E-03	-5.9767E-03	1.9479E-03	-2.4942E-03
165	-2.1981E+02	-1.4724E+01	-2.0509E+02	-5.4952E+01	3.3201E-04	-3.7853E-03	-4.5583E-03	-4.6491E-03	1.1958E-03	-1.8890E-03
170	-1.6046E+02	-4.8385E+00	-1.5562E+02	-2.7441E+01	-3.2229E-04	-2.8877E-03	-2.2762E-03	-3.1801E-03	6.1475E-04	-1.0411E-03
175	-8.5256E+01	-6.4761E-01	-8.4609E+01	-7.4023E+00	1.9724E-04	-1.5747E-03	-6.1403E-04	-1.6146E-03	2.3709E-04	-2.9700E-04
180	3.2524E-04	3.4532E-17	3.2524E-04	-1.0598E-10	-7.8584E-10	6.0594E-09	-8.7909E-15	6.0594E-09	-7.8584E-10	-4.3299E-15

STRAINS IN PLIES

THT	ANGULO CAPA 0.	ANGULO CAPA 90.	ANGULO CAPA 45.	ANGULO CAPA -45.	RF 1	RF 2	RF 3	RF 4	MIN RF
0	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	1000.00	1000.00	1000.00	1000.00	1000.00
5	-1.9724E-04	1.5747E-03	3.8174E-04	9.9577E-04	33.80	8.47	34.93	13.39	8.47
10	-3.2229E-04	2.8877E-03	1.4458E-04	2.4208E-03	20.69	4.62	92.22	5.51	4.62
15	-3.3201E-04	3.7853E-03	-5.5255E-04	4.0058E-03	20.08	3.52	12.07	3.33	3.33
20	-2.1926E-04	4.2481E-03	-1.4879E-03	5.5167E-03	30.41	3.14	4.48	2.42	2.42
25	-3.6473E-07	4.3440E-03	-2.4685E-03	6.8122E-03	1000.00	3.07	2.70	1.96	1.96
30	3.0065E-04	4.1692E-03	-3.3658E-03	7.8356E-03	44.35	3.20	1.98	1.70	1.70
35	6.6163E-04	3.8115E-03	-4.1082E-03	8.5814E-03	20.15	3.50	1.62	1.55	1.55
40	1.0653E-03	3.3381E-03	-4.6605E-03	9.0638E-03	12.52	3.99	1.43	1.47	1.43
45	1.4981E-03	2.7957E-03	-5.0043E-03	9.2981E-03	8.90	4.77	1.33	1.43	1.33
50	1.9476E-03	2.2163E-03	-5.1258E-03	9.2897E-03	6.85	6.02	1.30	1.44	1.30
55	2.3972E-03	1.6235E-03	-5.0076E-03	9.0283E-03	5.56	8.21	1.33	1.48	1.33
60	2.8199E-03	1.0388E-03	-4.6268E-03	8.4855E-03	4.73	12.84	1.44	1.57	1.44
65	3.1665E-03	4.8934E-04	-3.9608E-03	7.6166E-03	4.21	27.25	1.68	1.75	1.68
70	3.3515E-03	1.6122E-05	-3.9074E-03	6.3749E-03	3.98	827.02	2.22	2.09	2.09
75	3.2404E-03	-3.1972E-04	-1.8340E-03	4.7546E-03	4.11	20.85	3.64	2.80	2.80
80	2.6688E-03	-4.4447E-04	-6.5738E-04	2.8817E-03	5.00	15.00	10.14	4.63	4.63
85	1.5411E-03	-3.1605E-04	1.0812E-04	1.1169E-03	8.65	21.09	123.32	11.94	8.65
90	-3.0297E-09	6.5920E-10	-1.1853E-09	-1.1853E-09	1000.00	1000.00	1000.00	1000.00	1000.00
95	-1.5411E-03	3.1605E-04	-1.1170E-03	-1.0811E-04	4.33	42.19	5.97	61.67	4.33
100	-2.6688E-03	4.4447E-04	-2.8817E-03	6.5738E-04	2.50	30.00	2.31	20.28	2.31
105	-3.2404E-03	3.1972E-04	-4.7546E-03	1.8340E-03	2.06	41.70	1.40	7.27	1.40

110	-3.3515E-03	-1.6123E-05	-6.3749E-03	3.0074E-03	1.99	413.50	1.05	4.43	1.05
115	-3.1665E-03	-4.8934E-04	-7.6166E-03	3.9608E-03	2.11	13.62	0.88	3.37	0.88
120	-2.8199E-03	-1.0388E-03	-8.4855E-03	4.6268E-03	2.36	6.42	0.79	2.88	0.79
125	-2.3972E-03	-1.6235E-03	-9.0283E-03	5.0076E-03	2.78	4.11	0.74	2.66	0.74
130	-1.9476E-03	-2.2163E-03	-9.2897E-03	5.1258E-03	3.42	3.01	0.72	2.60	0.72
135	-1.4981E-03	-2.7957E-03	-9.2981E-03	5.0043E-03	4.45	2.38	0.72	2.66	0.72
140	-1.0653E-03	-3.3381E-03	-9.0638E-03	4.6605E-03	6.26	2.00	0.74	2.86	0.74
145	-6.6163E-04	-3.8115E-03	-8.5814E-03	4.1082E-03	10.08	1.75	0.78	3.25	0.78
150	-3.0065E-04	-4.1692E-03	-7.8356E-03	3.3658E-03	22.18	1.60	0.85	3.96	0.85
155	3.6575E-07	-4.3440E-03	-6.8121E-03	2.4685E-03	1000.00	1.53	0.98	5.40	0.98
160	2.1926E-04	-4.2481E-03	-5.5167E-03	1.4879E-03	60.81	1.57	1.21	8.96	1.21
165	3.3201E-04	-3.7853E-03	-4.0058E-03	5.5254E-04	40.16	1.76	1.66	24.13	1.66
170	3.2229E-04	-2.8877E-03	-2.4208E-03	-1.4458E-04	41.37	2.31	2.75	46.11	2.31
175	1.9724E-04	-1.5747E-03	-9.9576E-04	-3.8174E-04	67.60	4.23	6.70	17.46	4.23
180	-7.8584E-10	6.0594E-09	2.6368E-09	2.6368E-09	1000.00	1000.00	1000.00	1000.00	1000.00

MINIMUM RESERVE FACTOR = 0.72 THETA = 45.

Output results graph:

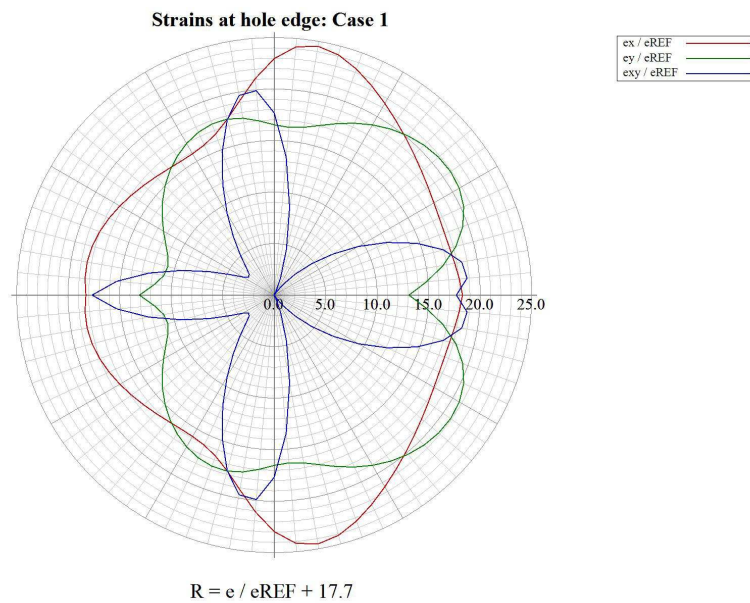


Figure 5: Ejemplo 1 Results graph: Cartesian Global Axis

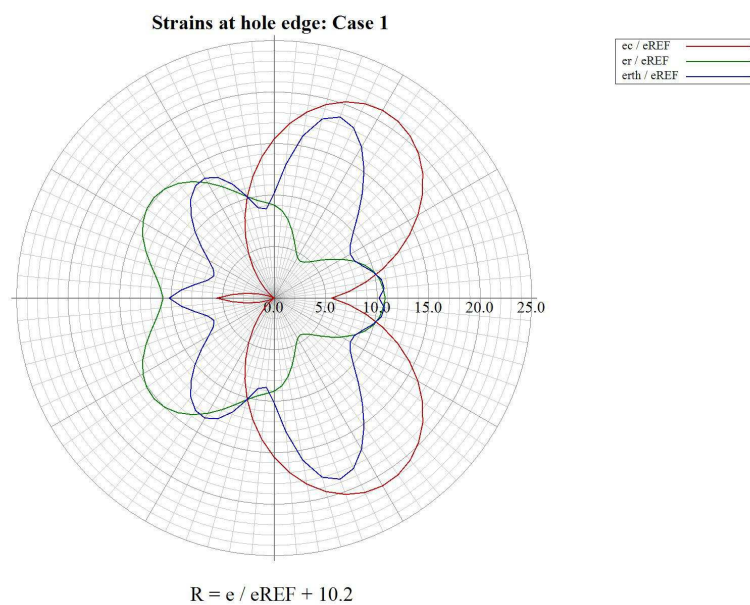


Figure 6: Ejemplo 1 Results graph: Polar Global Axis